UC Davis FCSR - Aurora User Guide

Start Up

1. Login to Windows:
    a. Username: My Aurora
    b. Password: Welcome#1
    c. The computer should always be left on, but the power button for the monitor itself found is on the bottom right corner.
2. Check the fluidics
    a. Waste should have 1L bleach and be less than half full when you start.
    b. Sheath (DI Water) should be over ⅓ full.
        i. When changing the sheath be careful not to touch the probe to anything other than the probe holder. Ethanol your gloves and the holder before and after changing the box, to ensure cleanliness.
3. Turn on the Aurora, open SpectroFlo software and login
    a. The power button is located on the left hand side of the instrument.
    b. Login with your individual or lab's login.
4. Cytometer warm up and cleaning
    a. Open the Acquisition menu.
    b. The Aurora will go through its startup routine checking valves, purging the sheath filter, and calibrating the SIT height. Once this is completed, the "Cytometer" indicator in the bottom right corner of the screen will turn green.
    c. The optical bench needs 45 minutes to warm up prior to QC.
    d. While the optical bench warms up, soak the flow cell with 30% Contrad for 30 minutes.
        i. Select the "Cytometer" menu on the left side of the screen (bottom button.)
        ii. Click "Clean Flow Cell."
        iii. Load 3mL 30% Contrad onto the sample port and click "Continue."
        iv. Let the flow cell soak in Contrad for 30 minutes before proceeding.
        v. After 30 minutes, load a tube of 3mL DI water onto the sample port and click "Continue."
    e. Clean system with DI water.
        i. Go to the "Experiment" menu on the left side of the screen (top button) and open the default experiment.
        ii. Fill a FACS tube with filtered DI water to within ½-inch of the top of the tube.
        iii. Hold the FACS tube up to the sample port, but DO NOT click it into place. Water should surround the bottom part of the metal SIP, but do not let it get into the bottom of the loading port.
        iv. While holding the FACS tube, select "SIT Flush" from the Acquisition control panel. Water will be sucked into the SIT. Repeat this for a total of three SIT Flushes.

v. Run filtered DI water for 10 to 15 minutes on high. You are looking for the event rate to fall below 10 events/second.
vi. Click stop and proceed to QC & Setup

5. Run QC & Setup
    a. Navigate to the QC& Setup Menu in the blue ribbon at the top right corner of the screen.
    b. QC beads are located in a labeled box next door in the cold room.
        i. If there is already a prepared tube in the box and the date is within a week, you can use it for QC. Otherwise, prepare new beads.
    c. Preparing QC Beads:
        i. Write the bead lot and date on a new tube.
        ii. Using the syringe filtered DI water, add 300-500uL of water
        iii. Vortex bead bottle well and then squeeze 1 full drop into the tube and vortex.
    d. Load the QC beads onto the sample port.
    e. Select the correct bead lot from the drop down menu and press start.
        i. Check to see that the event rate is above 100 events/sec. If it is not, QC is likely to fail. If the event rate is too low, click abort and either add another drop of beads to the freshly made tube or make a new tube of beads if you were reusing an old tube.

Experiment Setup

1. Return to the Acquisition Menu
2. Click “New” and a setup wizard will guide you through the process.
3. Enter an experiment name.
4. Select all fluorescent tags used in the experiment.
    a. Start typing the names into the search box on the top right corner to start to filter the options in the panel below. Fluorescent tags are grouped by the primary exciting laser.
    b. Double click the appropriate tag to add it to the selected list to the right.
    c. Check that you have added all of your tags and click “Next.”
5. Create your Reference and Experimental Groups
    a. Select “Reference Group” to create a group for tubes that will be used as reference controls.
        i. Select the control type (cells or beads) for the unstained control. The unstained control needs to be of the same type and prepared in the same way as the experimental samples.
        ii. Select the “Control Type” (beads or cells) for each single color control.
        iii. Add the “Label” for each control (i.e. CD3, TNFa, Viability etc.)
        iv. Enter Lot information for your reagents (recommended, but optional)
        v. *Optional:* if using a universal negative, assign the appropriate type to each single color control.

1. If needed, add additional negative controls by checking the "Define Universal Negative Control(s)" at the top of the window.

vi. Click "Save"

b. Create groups for your samples and add tubes by selecting "Group" and/or "Tube" buttons.
   i. FCS files will be named with the "Tube" name only.
   ii. Click next when finished.

6. In the "Markers" tab, type in labels for each fluorescent tag.
   a. This can be done at the experiment or group level.
   b. Don't skip this step as this is what will provide axis labels in FlowJo.
   c. Click next
7. Set up Acquisition parameters
   a. Select the "Experiment User Setting"
      i. If this is a new experiment, use the default "Cytek Assay Settings (Admin)."
      ii. If you have previously defined assay settings, select those.
   b. Select the acquisition worksheet
      i. Default: "Default Raw Worksheet (Raw)
      ii. Popular: HyperCompact v3 (Raw) [puts all histograms on the screen without needing to scroll.]
   c. Select the stopping gate (ex: All events or P1)
   d. Select storage gate (**in almost all cases, this should be "All Events."**)
      i. This is what data will be recorded in your FCS file, anything not in this gate will not be recorded!
      ii. Please check with the Flow Core staff before you select anything other than "All Events" for the storage gate.
   e. Select the number of events in the stopping gate to record.
      i. Alternatively, if desired, you can stop acquisition by volume or time.
   f. Click next.
8. Click Save and Open to open the new experiment.

Sample Acquisition & Data Export

1. Define Instrument Settings
   a. Select your unstained sample in the reference control group. Vortex and load sample onto the sample port.
   b. In the Acquisition Control panel, with flow rate on "low" start acquiring data, do not record yet.
   c. In the Instrument Control Panel:
      i. In the "Signal" tab, select which area, height and width parameters to record
      ii. In the "Gain" tab, adjust FSC and SSC gains to put your cells on scale.
      iii. The threshold can be adjusted in the "Threshold" tab as needed.
      iv. Stop acquisition.

v. Check all single color controls to make sure their signals are on scale.
    - *Note:* **Do not change gains for individual fluorescent detectors.** The relationships between each detector must be maintained to produce appropriate spectral signatures.
    - If your fluorescent signals are off scale, it is recommended that you titrate your reagents (especially viability dyes.)
    - If you have a fluorescent protein that cannot be titrated, then use the "All Channels %" to decrease the gains by the same percentage across all detectors.

vi. Once you have your instrument control settings set, you can click the "Save As" button to save the settings for future experiments.

2. Acquire and record all reference control group samples at the same rate that you plan on running your experiment samples. Best practice is to run everything on "low."
3. Perform Spectral Unmixing.
    a. Once you have acquired all reference control group samples, the "Unmix" button in the Experiment window will become live. Click "Unmix" to open the wizard.
    b. Check that all unstained and single color control samples are listed.
    c. In the bottom right corner select the "Unmixing Model" either with or without AF (autofluorescence) extraction.
    d. For each control:
        i. Place the P1 gate around the appropriate population (cells or beads.)
        ii. Define the positive and negative populations.
            - Aim for at least 1,000-2,000 positive events. Record more events if needed.
            - Gate on the "brightest of the bright" population. This will tighten up the rainbow boxes in the spectral signature. The lower the variation in your signal, the better the unmixing calculation.
            - Confirm the spectral signatures are correct for the fluorescent tag selected.
    e. When finished, select "Live Unmix" to return to the experiment.
4. Acquire and record all experimental samples.
    a. After unmixing, you can create an unmixed worksheet to view the unmixed data during acquisition. Use the tools in the ribbon on top of the worksheet to create a gating strategy.
5. Save any raw or unmixed worksheets for future experiments using the disk icons in the worksheet ribbon.
6. Click Save in the experiment window and close out of the experiment.
7. To export your data:
    a. Select "My Experiments" in the Acquisition menu
    b. Click on the experiment name and select export.
    c. Export your data first to the computer: Data (D:) > User Data and Exports > "Your Lab Folder."
    d. Use a flash drive, external hard drive or UC Davis Box to transfer your data back to your lab.

Shutdown

1. Using the left hand toolbar, select “Cytometer” then "Fluidics Shutdown” and follow the prompts. Make sure the cleaning solutions are not touching the metal SIP as this can cause issues with QC.
    a. 3mL of 10% bleach
    b. 3mL DI water
    c. 3mL 30% contrad
    d. 3mL DI water
2. Make sure there is water in the tube covering the bottom of the sample line
3. Turn off the cytometer.
4. Exit SpectroFlo.
5. Please leave the computer on. The FCSR has an automated backup system for the cytometer database and user data that occurs on nights and weekends.

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